MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) : SEMESTER IV

DSE: MG4DSEPHY200: SEMICONDUCTOR ELECTRONICS

Duration: 1.5 hours **Maximum Marks: 50**

[Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)]

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any 10 questions: Each question carries 2 marks

1. What is meant by the depletion region in a JFET? [U] / [1]
2. Name the regions seen in the drain characteristics of a JFET. [U] / [1]
3. Write any two advantages and disadvantages of IC'S. [K] / [3]
4. Draw the block diagram of a typical op-amp. [K] / [4]
5. What is meant by transfer characteristics of a depletion MOSFET? [U] / [2]
6. Write the expression for the voltage gain of a Non-Inverting Amplifier and identify the terms [K] / [4]
7. Draw the circuit diagram of a three-input inverting summing amplifier. [U] / [4]
8. How does an op-amp work as a comparator? [U] / [4]
9. Give the symbol and truth table of AND gate with two inputs [U] / [5]
10. Convert 40.625_{10} into binary equivalent [U] / [5]
11. Add 110011_2 to 101101_2 [U] / [5]
12. State any two laws of Boolean algebra. [K] / [6]
13. What is meant by standard forms of Boolean expressions? [U] / [6]
14. Draw the truth table for the Boolean expression Y=A+BC [A] / [6]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any 6 questions: Each question carries 5 marks

15. Explain the construction and working of a JFET. [U] / [1]

16. Discuss the transfer characteristics of an enhancement-type MOSFET. [U] / [2]

17. State and explain De Morgan's theorems. [U] / [6]

18. Describe the fabrication of resistors, capacitors, and transistors in a monolithic IC.

[U] / [3]

19. Explain the working of an op-amp integrator with a neat circuit diagram and derive the expression for its output voltage. [U] / [4]

20. With the help of a circuit diagram, explain how a unity gain buffer eliminates the loading effect in electronic circuits. [A] / [4]

21. Subtract 1010_2 from 1101_2 using 1's complement method. [A] / [5]

22. Simply the Boolean expression $A\left[B + C\left(\overline{AB + AC}\right)\right]$
[A] / [6]

[5x6 = 30]